

Erdős Problem #693

EP-693

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $k \geq 2$ and n be sufficiently large depending on k . Let $A = \{a_1 < a_2 < \dots\}$ be the set of those integers in $[n, n^k]$ which have a divisor in $(n, 2n)$. Estimate $\max_i a_{i+1} - a_i$. Is this $\leq (\log n)^{O(1)}$?

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”difficulty”: ”L1”

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>